Write your name here

Surname	Other names

Pearson Edexcel
Level 3 GCE

Centre Number | Candidate Number

Mathematics

Advanced Subsidiary
Paper 2: Statistics and Mechanics

Wednesday 23 May 2018 – Morning
Time: 1 hour 15 minutes

Paper Reference
8MA0/02

You must have:
Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- There are **two** sections in this question paper. Answer **all** the questions in Section A and **all** the questions in Section B.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 9 questions in this question paper. The total mark for this paper is 60.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

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SECTION A: STATISTICS

Answer ALL questions. Write your answers in the spaces provided.

1. A company is introducing a job evaluation scheme. Points (x) will be awarded to each job based on the qualifications and skills needed and the level of responsibility. Pay (£y) will then be allocated to each job according to the number of points awarded.

Before the scheme is introduced, a random sample of 8 employees was taken and the linear regression equation of pay on points was $y = 4.5x - 47$

(a) Describe the correlation between points and pay.
(1)

(b) Give an interpretation of the gradient of this regression line.
(1)

(c) Explain why this model might not be appropriate for all jobs in the company.
(1)

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Question 1 continued

(Total for Question 1 is 3 marks)

2. A factory buys 10% of its components from supplier A, 30% from supplier B and the rest from supplier C. It is known that 6% of the components it buys are faulty.

Of the components bought from supplier A, 9% are faulty and of the components bought from supplier B, 3% are faulty.

(a) Find the percentage of components bought from supplier C that are faulty.

(3)

A component is selected at random.

(b) Explain why the event "the component was bought from supplier B" is not statistically independent from the event "the component is faulty".

(1)

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Question 2 continued

(Total for Question 2 is 4 marks)

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3. Naasir is playing a game with two friends. The game is designed to be a game of chance so that the probability of Naasir winning each game is $\frac{1}{3}$

Naasir and his friends play the game 15 times.

(a) Find the probability that Naasir wins

(i) exactly 2 games,

(ii) more than 5 games.

(3)

Naasir claims he has a method to help him win more than $\frac{1}{3}$ of the games. To test this claim, the three of them played the game again 32 times and Naasir won 16 of these games.

(b) Stating your hypotheses clearly, test Naasir's claim at the 5% level of significance.

(4)

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Question 3 continued

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Question 3 continued

(Total for Question 3 is 7 marks)

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4. Helen is studying the daily mean wind speed for Camborne using the large data set from 1987. The data for one month are summarised in Table 1 below.

Windspeed	n/a	6	7	8	9	11	12	13	14	16
Frequency	13	2	3	2	2	3	1	2	1	2

Table 1

(a) Calculate the mean for these data.

(1)

(b) Calculate the standard deviation for these data and state the units.

(2)

The means and standard deviations of the daily mean wind speed for the other months from the large data set for Camborne in 1987 are given in Table 2 below. The data are not in month order.

Month	*A*	*B*	*C*	*D*	*E*
Mean	7.58	8.26	8.57	8.57	11.57
Standard Deviation	2.93	3.89	3.46	3.87	4.64

Table 2

(c) Using your knowledge of the large data set, suggest, giving a reason, which month had a mean of 11.57

(2)

The data for these months are summarised in the box plots on the opposite page. They are not in month order or the same order as in Table 2.

(d) (i) State the meaning of the * symbol on some of the box plots.

(ii) Suggest, giving your reasons, which of the months in Table 2 is most likely to be summarised in the box plot marked *Y*.

(3)

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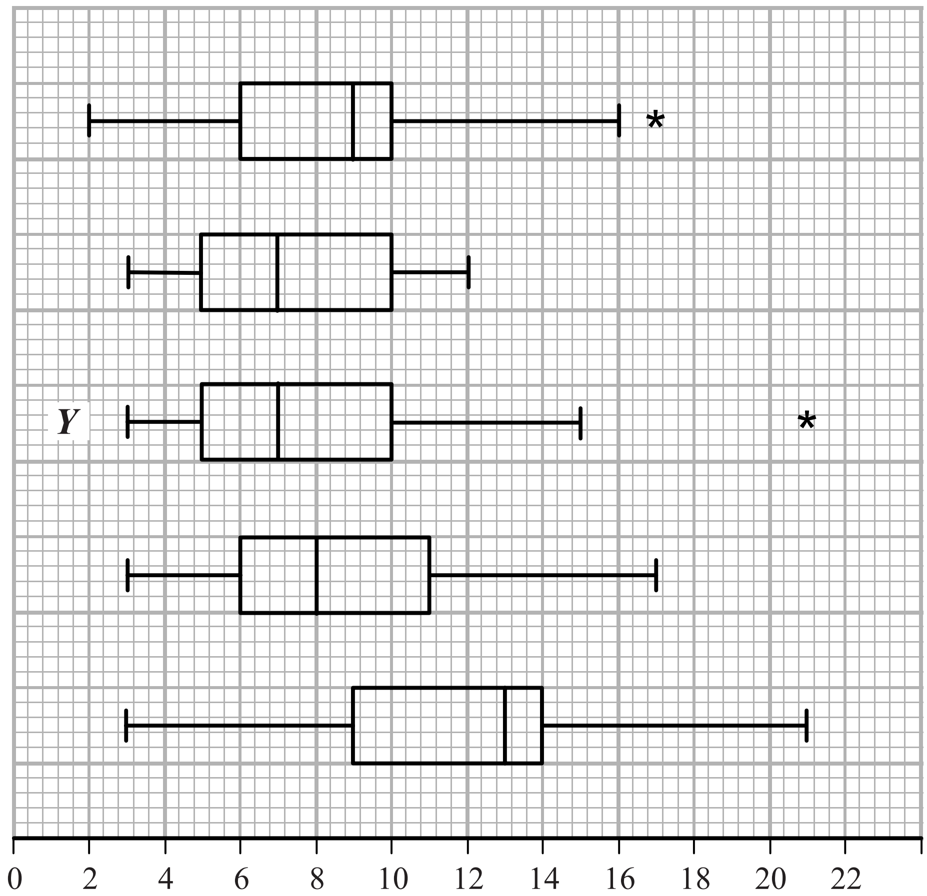
Question 4 continued



(Total for Question 4 is 8 marks)

5. A biased spinner can only land on one of the numbers 1, 2, 3 or 4. The random variable X represents the number that the spinner lands on after a single spin and $P(X = r) = P(X = r + 2)$ for $r = 1, 2$

Given that $P(X = 2) = 0.35$

(a) find the complete probability distribution of X.

(2)

Ambroh spins the spinner 60 times.

(b) Find the probability that more than half of the spins land on the number 4
Give your answer to 3 significant figures.

(3)

The random variable $Y = \dfrac{12}{X}$

(c) Find $P(Y - X \leqslant 4)$

(3)

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 8 marks)

TOTAL FOR SECTION A IS 30 MARKS

Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number | Candidate Number

Wednesday 22 May 2019

Morning | Paper Reference **8MA0-21**

Mathematics

Advanced Subsidiary
Paper 21: Statistics

You must have:
Mathematical Formulae and Statistical Tables, calculator

Total Marks

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- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 30. There are 5 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

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Turn over

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Answer ALL questions. Write your answers in the spaces provided.

1. A sixth form college has 84 students in Year 12 and 56 students in Year 13

The head teacher selects a stratified sample of 40 students, stratified by year group.

(a) Describe how this sample could be taken.

(3)

The head teacher is investigating the relationship between the amount of sleep, s hours, that each student had the night before they took an aptitude test and their performance in the test, p marks.
For the sample of 40 students, he finds the equation of the regression line of p on s to be

$$p = 26.1 + 5.60s$$

(b) With reference to this equation, describe the effect that an extra 0.5 hours of sleep may have, on average, on a student's performance in the aptitude test.

(1)

(c) Describe one limitation of this regression model.

(1)

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Question 1 continued

(Total for Question 1 is 5 marks)

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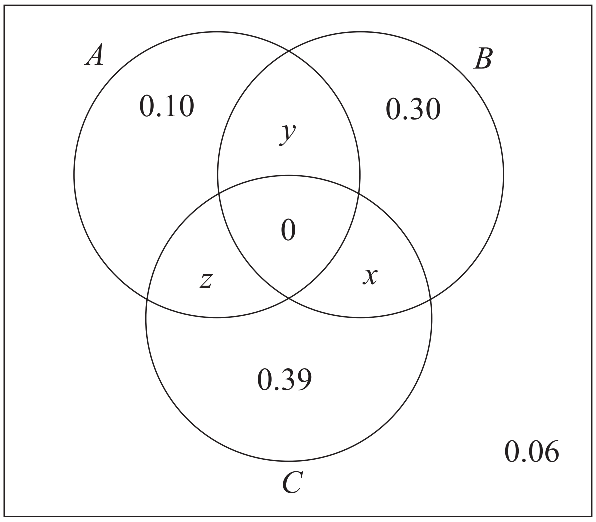
2. The Venn diagram shows three events, A, B and C, and their associated probabilities.



Events B and C are mutually exclusive.
Events A and C are independent.

Showing your working, find the value of x, the value of y and the value of z.

(5)

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Question 2 continued

(Total for Question 2 is 5 marks)

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3. A fair 5-sided spinner has sides numbered 1, 2, 3, 4 and 5

The spinner is spun once and the score of the side it lands on is recorded.

(a) Write down the name of the distribution that can be used to model the score of the side it lands on.

(1)

The spinner is spun 28 times.

The random variable X represents the number of times the spinner lands on 2

(b) (i) Find the probability that the spinner lands on 2 at least 7 times.

(ii) Find $P(4 \leqslant X < 8)$

(5)

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Question 3 continued

(Total for Question 3 is 6 marks)

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4. Joshua is investigating the daily total rainfall in Hurn for May to October 2015

Using the information from the large data set, Joshua wishes to calculate the mean of the daily total rainfall in Hurn for May to October 2015

(a) Using your knowledge of the large data set, explain why Joshua needs to clean the data before calculating the mean.

(1)

Using the information from the large data set, he produces the grouped frequency table below.

Daily total rainfall (r mm)	Frequency	Midpoint (x mm)
$0 \leqslant r < 0.5$	121	0.25
$0.5 \leqslant r < 1.0$	10	0.75
$1.0 \leqslant r < 5.0$	24	3.0
$5.0 \leqslant r < 10.0$	12	7.5
$10.0 \leqslant r < 30.0$	17	20.0

You may use $\sum \mathrm{f}x = 539.75$ and $\sum \mathrm{f}x^2 = 7704.1875$

(b) Use linear interpolation to calculate an estimate for the upper quartile of the daily total rainfall.

(2)

(c) Calculate an estimate for the standard deviation of the daily total rainfall in Hurn for May to October 2015

(2)

(d) (i) State the assumption involved with using class midpoints to calculate an estimate of a mean from a grouped frequency table.

(ii) Using your knowledge of the large data set, explain why this assumption does not hold in this case.

(iii) State, giving a reason, whether you would expect the actual mean daily total rainfall in Hurn for May to October 2015 to be larger than, smaller than or the same as an estimate based on the grouped frequency table.

(3)

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Question 4 continued

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Question 4 continued

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Question 4 continued

(Total for Question 4 is 8 marks)

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5. Past records show that 15% of customers at a shop buy chocolate. The shopkeeper believes that moving the chocolate closer to the till will increase the proportion of customers buying chocolate.

After moving the chocolate closer to the till, a random sample of 30 customers is taken and 8 of them are found to have bought chocolate.

Julie carries out a hypothesis test, at the 5% level of significance, to test the shopkeeper's belief.

Julie's hypothesis test is shown below.

$H_0 : p = 0.15$

$H_1 : p \geqslant 0.15$

Let X = the number of customers who buy chocolate.

$X \sim B(30, 0.15)$

$P(X = 8) = 0.0420$

$0.0420 < 0.05$ so reject H_0

There is sufficient evidence to suggest that the proportion of customers buying chocolate has increased.

(a) Identify the first two errors that Julie has made in her hypothesis test.
(2)

(b) Explain whether or not these errors will affect the conclusion of her hypothesis test. Give a reason for your answer.
(1)

(c) Find, using a 5% level of significance, the critical region for a one-tailed test of the shopkeeper's belief. The probability in the tail should be less than 0.05
(2)

(d) Find the actual level of significance of this test.
(1)

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 6 marks)

TOTAL FOR STATISTICS IS 30 MARKS

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Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number | Candidate Number

Wednesday 14 October 2020

Afternoon | Paper Reference **8MA0/21**

Mathematics

Advanced Subsidiary
Paper 21: Statistics

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Values from statistical tables should be quoted in full. If a calculator is used instead of tables the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 30. There are 5 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question*.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over

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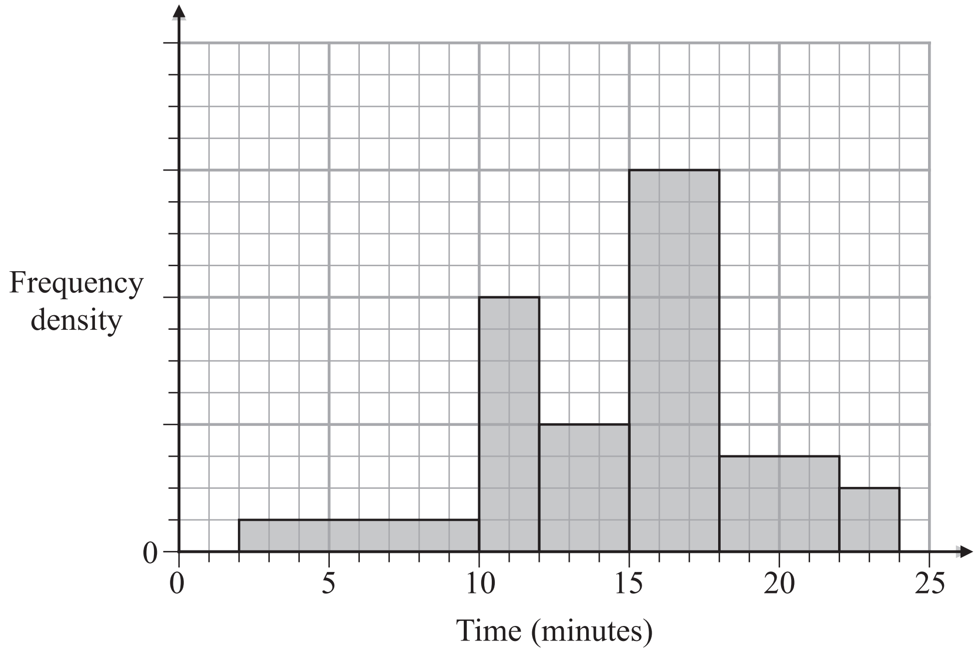

Figure 1

The histogram in Figure 1 shows the times taken to complete a crossword by a random sample of students.

The number of students who completed the crossword in more than 15 minutes is 78

Estimate the percentage of students who took less than 11 minutes to complete the crossword.

(4)

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Question 1 continued

(Total for Question 1 is 4 marks)

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2. Jerry is studying visibility for Camborne using the large data set June 1987.

The table below contains two extracts from the large data set.

It shows the daily maximum relative humidity and the daily mean visibility.

Date	**Daily Maximum Relative Humidity**	**Daily Mean Visibility**
Units	**%**	
10/06/1987	90	5300
28/06/1987	100	0

(The units for Daily Mean Visibility are deliberately omitted.)

Given that daily mean visibility is given to the nearest 100,

(a) write down the range of distances in metres that corresponds to the recorded value 0 for the daily mean visibility.

(1)

Jerry drew the following scatter diagram, Figure 2, and calculated some statistics using the June 1987 data for Camborne from the large data set.

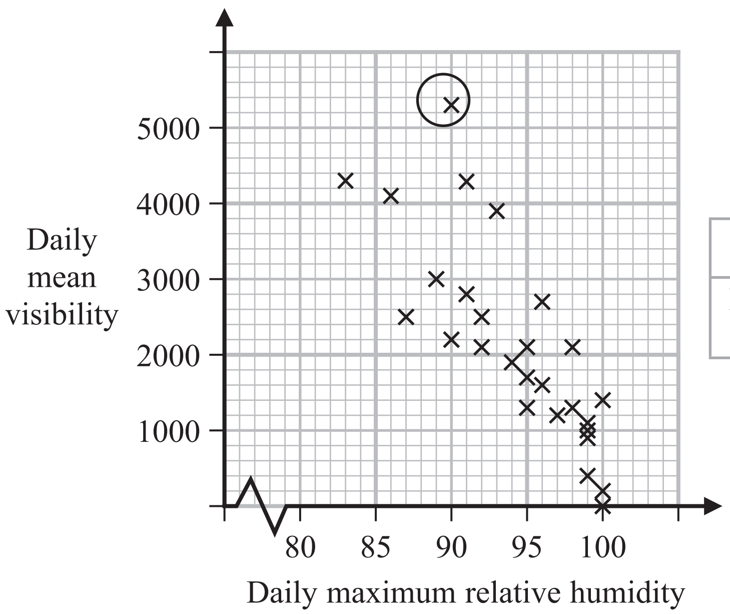

Figure 2

	Q_1	IQR
Daily mean visibility	1100	1600
Daily maximum relative humidity (%)	92	8

Jerry defines an outlier as a value that is more than 1.5 times the interquartile range above Q_3 or more than 1.5 times the interquartile range below Q_1.

(b) Show that the point circled on the scatter diagram is an outlier for visibility.

(2)

(c) Interpret the correlation between the daily mean visibility and the daily maximum relative humidity.

(1)

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Jerry drew the following scatter diagram, Figure 3, using the June 1987 data for Camborne from the large data set, but forgot to label the *x*–axis.

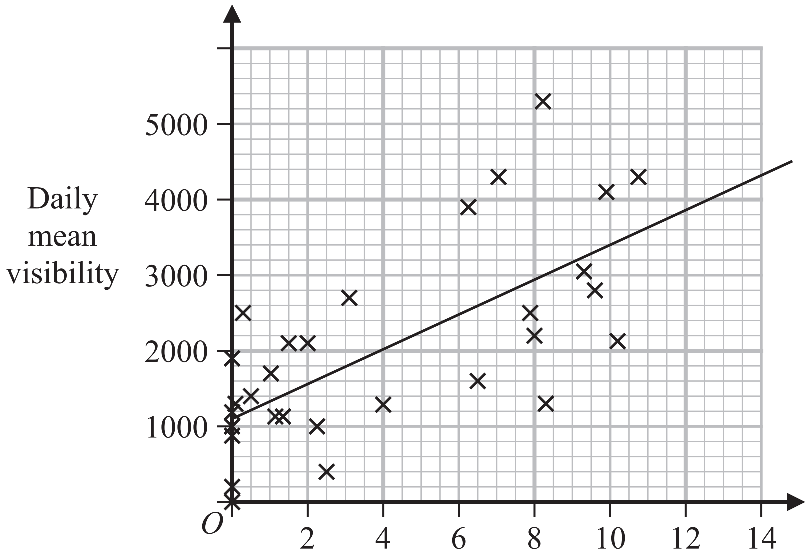

Figure 3

(d) Using your knowledge of the large data set, suggest which variable the *x*-axis on this scatter diagram represents.

(1)

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Question 2 continued

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Question 2 continued

(Total for Question 2 is 5 marks)

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3. In a game, a player can score 0, 1, 2, 3 or 4 points each time the game is played.

The random variable S, representing the player's score, has the following probability distribution where a, b and c are constants.

s	0	1	2	3	4
$P(S = s)$	a	b	c	0.1	0.15

The probability of scoring less than 2 points is twice the probability of scoring at least 2 points.

Each game played is independent of previous games played.

John plays the game twice and adds the two scores together to get a total.

Calculate the probability that the total is 6 points.

(6)

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Question 3 continued

(Total for Question 3 is 6 marks)

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4. A lake contains three different types of carp.

There are an estimated 450 mirror carp, 300 leather carp and 850 common carp.

Tim wishes to investigate the health of the fish in the lake.

He decides to take a sample of 160 fish.

(a) Give a reason why stratified random sampling cannot be used.

(1)

(b) Explain how a sample of size 160 could be taken to ensure that the estimated populations of each type of carp are fairly represented.

You should state the name of the sampling method used.

(2)

As part of the health check, Tim weighed the fish.

His results are given in the table below.

Weight (w kg)	Frequency (f)	Midpoint (m kg)
$2 \leqslant w < 3.5$	8	2.75
$3.5 \leqslant w < 4$	32	3.75
$4 \leqslant w < 4.5$	64	4.25
$4.5 \leqslant w < 5$	40	4.75
$5 \leqslant w < 6$	16	5.5

(You may use $\sum \text{f}m = 692$ and $\sum \text{f}m^2 = 3053$)

(c) Calculate an estimate for the standard deviation of the weight of the carp.

(2)

Tim realised that he had transposed the figures for 2 of the weights of the fish.

He had recorded in the table 2.3 instead of 3.2 and 4.6 instead of 6.4

(d) Without calculating a new estimate for the standard deviation, state what effect

(i) using the correct figure of 3.2 instead of 2.3

(ii) using the correct figure of 6.4 instead of 4.6

would have on your estimated standard deviation.

Give a reason for each of your answers.

(2)

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Question 4 continued

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Question 4 continued

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Question 4 continued

(Total for Question 4 is 7 marks)

5. Afrika works in a call centre.

She assumes that calls are independent and knows, from past experience, that on each sales call that she makes there is a probability of $\frac{1}{6}$ that it is successful.

Afrika makes 9 sales calls.

(a) Calculate the probability that at least 3 of these sales calls will be successful.

(2)

The probability of Afrika making a successful sales call is the same each day.

Afrika makes 9 sales calls on each of 5 different days.

(b) Calculate the probability that at least 3 of the sales calls will be successful on exactly 1 of these days.

(2)

Rowan works in the same call centre as Afrika and believes he is a more successful salesperson.

To check Rowan's belief, Afrika monitors the next 35 sales calls Rowan makes and finds that 11 of the sales calls are successful.

(c) Stating your hypotheses clearly test, at the 5% level of significance, whether or not there is evidence to support Rowan's belief.

(4)

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 8 marks)

TOTAL FOR STATISTICS IS 30 MARKS

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Pearson Edexcel Level 3 GCE

Centre Number

Candidate Number

Paper reference **8MA0/21**

Mathematics

Advanced Subsidiary
PAPER 21: Statistics

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Turn over ▶

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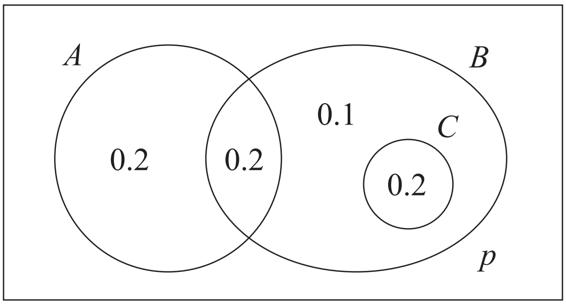

The Venn diagram, where p is a probability, shows the 3 events A, B and C with their associated probabilities.

(a) Find the value of p.

(1)

(b) Write down a pair of mutually exclusive events from A, B and C.

(1)

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Question 1 continued

(Total for Question 1 is 2 marks)

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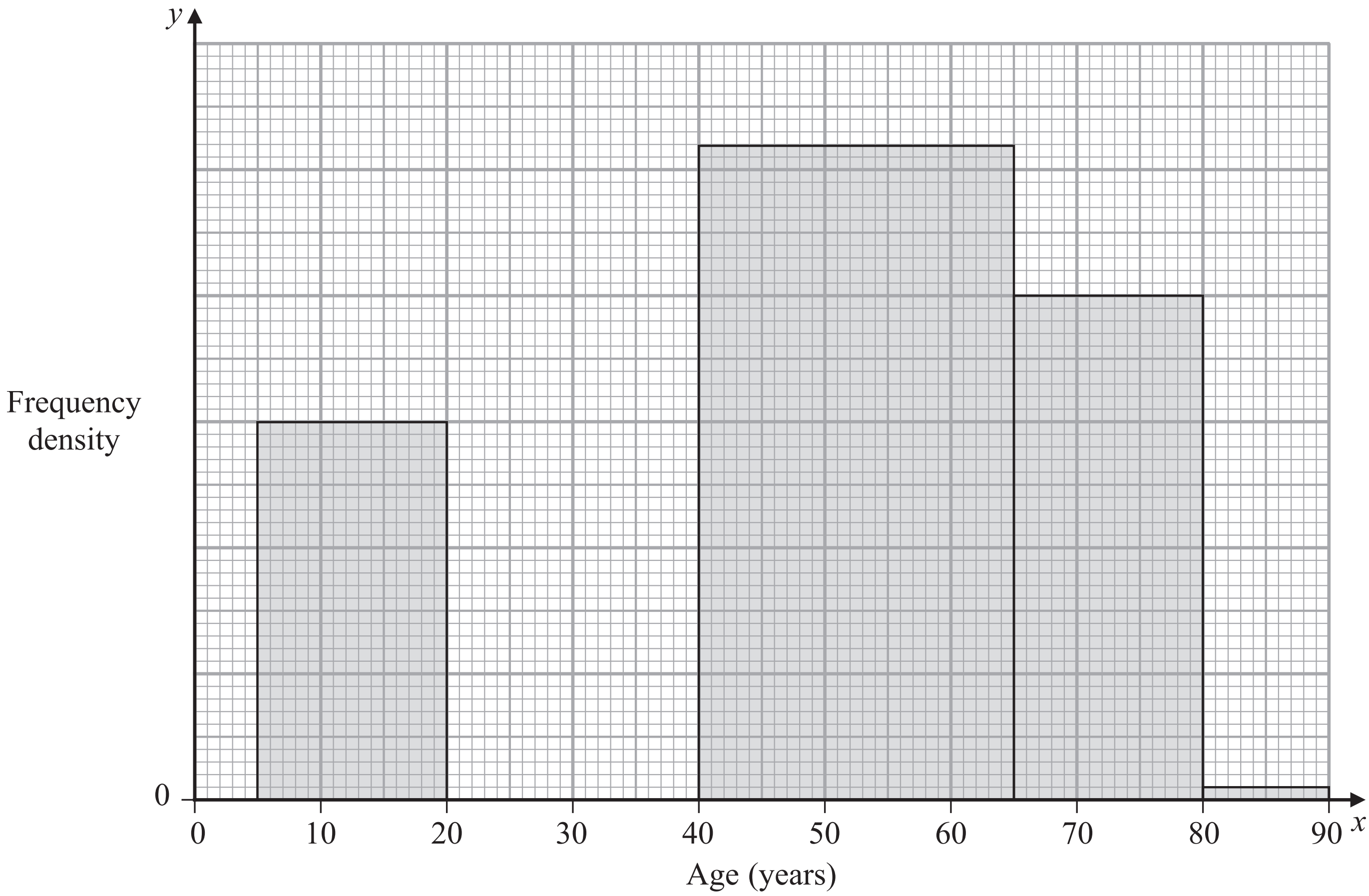
2. The partially completed table and partially completed histogram give information about the ages of passengers on an airline.

There were no passengers aged 90 or over.

Age (x years)	$0 \leqslant x < 5$	$5 \leqslant x < 20$	$20 \leqslant x < 40$	$40 \leqslant x < 65$	$65 \leqslant x < 80$	$80 \leqslant x < 90$
Frequency	5	45	90			1



(a) Complete the histogram.

(3)

(b) Use linear interpolation to estimate the median age.

(4)

An outlier is defined as a value greater than $Q_3 + 1.5 \times$ interquartile range.

Given that $Q_1 = 27.3$ and $Q_3 = 58.9$

(c) determine, giving a reason, whether or not the oldest passenger could be considered as an outlier.

(2)

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Question 2 continued

Question 2 continued

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Question 2 continued

(Total for Question 2 is 9 marks)

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3. Helen is studying one of the qualitative variables from the large data set for Heathrow from 2015.

She started with the data from 3rd May and then took every 10th reading.

There were only 3 different outcomes with the following frequencies

Outcome	*A*	*B*	*C*
Frequency	16	2	1

(a) State the sampling technique Helen used.

(1)

(b) From your knowledge of the large data set

(i) suggest which variable was being studied,

(ii) state the name of outcome *A*.

(2)

George is also studying the same variable from the large data set for Heathrow from 2015. He started with the data from 5th May and then took every 10th reading and obtained the following

Outcome	*A*	*B*	*C*
Frequency	16	1	1

Helen and George decided they should examine all of the data for this variable for Heathrow from 2015 and obtained the following

Outcome	*A*	*B*	*C*
Frequency	155	26	3

(c) State what inference Helen and George could reliably make from their original samples about the outcomes of this variable at Heathrow, for the period covered by the large data set in 2015.

(1)

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Question 3 continued

(Total for Question 3 is 4 marks)

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4. A nursery has a sack containing a large number of coloured beads of which 14% are coloured red.

Aliya takes a random sample of 18 beads from the sack to make a bracelet.

(a) State a suitable binomial distribution to model the number of red beads in Aliya's bracelet.

(1)

(b) Use this binomial distribution to find the probability that

(i) Aliya has just 1 red bead in her bracelet,

(ii) there are at least 4 red beads in Aliya's bracelet.

(3)

(c) Comment on the suitability of a binomial distribution to model this situation.

(1)

After several children have used beads from the sack, the nursery teacher decides to test whether or not the proportion of red beads in the sack has changed.
She takes a random sample of 75 beads and finds 4 red beads.

(d) Stating your hypotheses clearly, use a 5% significance level to carry out a suitable test for the teacher.

(4)

(e) Find the *p*-value in this case.

(1)

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Question 4 continued

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Question 4 continued

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Question 4 continued

(Total for Question 4 is 10 marks)

5. Two bags, **A** and **B**, each contain balls which are either red or yellow or green.

Bag **A** contains 4 red, 3 yellow and n green balls.
Bag **B** contains 5 red, 3 yellow and 1 green ball.

A ball is selected at random from bag **A** and placed into bag **B**.
A ball is then selected at random from bag **B** and placed into bag **A**.

The probability that bag **A** now contains an equal number of red, yellow and green balls is p.

Given that $p > 0$, find the possible values of n and p.

(5)

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 5 marks)

TOTAL FOR STATISTICS IS 30 MARKS

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Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Paper reference **8MA0/21**

Mathematics

Advanced Subsidiary
PAPER 21: Statistics

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Values from statistical tables should be quoted in full. If a calculator is used instead of tables the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 30. There are 5 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over

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1. The relationship between two variables p and t is modelled by the regression line with equation

$$p = 22 - 1.1\,t$$

The model is based on observations of the independent variable, t, between 1 and 10

(a) Describe the correlation between p and t implied by this model.
(1)

Given that p is measured in centimetres and t is measured in days,

(b) state the units of the gradient of the regression line.
(1)

Using the model,

(c) calculate the change in p over a 3-day period.
(2)

Tisam uses this model to estimate the value of p when $t = 19$

(d) Comment, giving a reason, on the reliability of this estimate.
(1)

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Question 1 continued

(Total for Question 1 is 5 marks)

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2. A manufacturer of sweets knows that 8% of the bags of sugar delivered from supplier A will be damp.
A random sample of 35 bags of sugar is taken from supplier A.

(a) Using a suitable model, find the probability that the number of bags of sugar that are damp is

(i) exactly 2

(ii) more than 3

(3)

Supplier B claims that when it supplies bags of sugar, the proportion of bags that are damp is less than 8%

The manufacturer takes a random sample of 70 bags of sugar from supplier B and finds that only 2 of the bags are damp.

(b) Carry out a suitable test to assess supplier B's claim.
You should state your hypotheses clearly and use a 10% level of significance.

(4)

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Question 2 continued

(Total for Question 2 is 7 marks)

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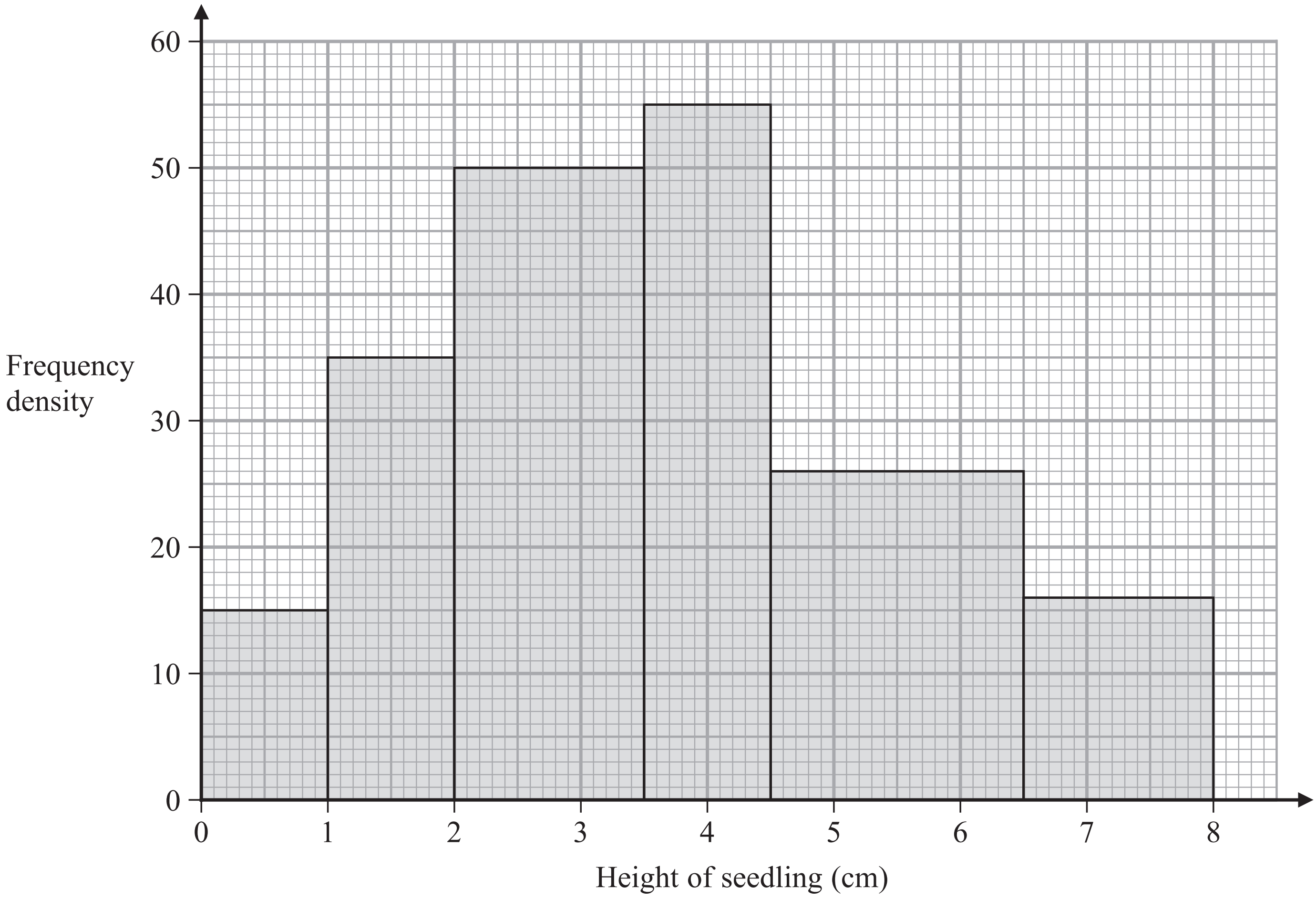
3. The histogram summarises the heights of 256 seedlings two weeks after they were planted.



(a) Use linear interpolation to estimate the median height of the seedlings.

(4)

Chris decides to model the **frequency density** for these 256 seedlings by a curve with equation

$$y = kx(8 - x) \qquad 0 \leqslant x \leqslant 8$$

where k is a constant.

(b) Find the value of k

(3)

Using this model,

(c) write down the median height of the seedlings.

(1)

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Question 3 continued

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Question 3 continued

(Total for Question 3 is 8 marks)

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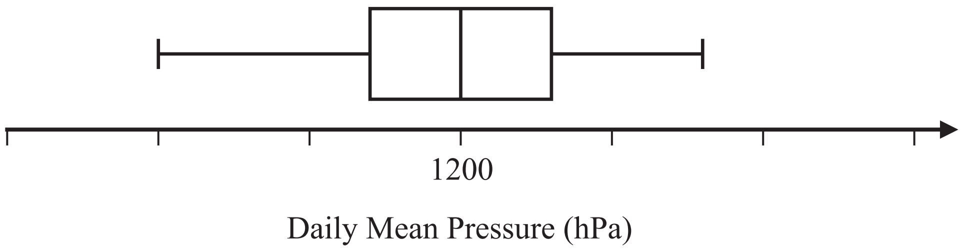
4. Jiang is studying the variable Daily Mean Pressure from the large data set.

He drew the following box and whisker plot for these data for one of the months for one location using a linear scale but

- he failed to label all the values on the scale
- he gave an incorrect value for the median



Using your knowledge of the large data set, suggest a suitable value for

(a) the median,

(1)

(b) the range.

(1)

(*You are not expected to have memorised values from the large data set. The question is simply looking for sensible answers*.)

(Total for Question 4 is 2 marks)

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5. Manon has two biased spinners, one red and one green.

The random variable R represents the score when the red spinner is spun.
The random variable G represents the score when the green spinner is spun.

The probability distributions for R and G are given below.

r	2	3
$\mathrm{P}(R = r)$	$\frac{1}{4}$	$\frac{3}{4}$

g	1	4
$\mathrm{P}(G = g)$	$\frac{2}{3}$	$\frac{1}{3}$

Manon spins each spinner once and adds the two scores.

(a) Find the probability that

(i) the sum of the two scores is 7

(ii) the sum of the two scores is less than 4

(3)

The random variable $X = mR + nG$ where m and n are integers.

$$\mathrm{P}(X = 20) = \frac{1}{6} \qquad \text{and} \qquad \mathrm{P}(X = 50) = \frac{1}{4}$$

(b) Find the value of m and the value of n

(5)

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 8 marks)

TOTAL FOR STATISTICS IS 30 MARKS

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Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Thursday 25 May 2023

Afternoon

Paper reference **8MA0/21**

Mathematics

Advanced Subsidiary
PAPER 21: Statistics

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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Information

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- The total mark for this part of the examination is 30. There are 5 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

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- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

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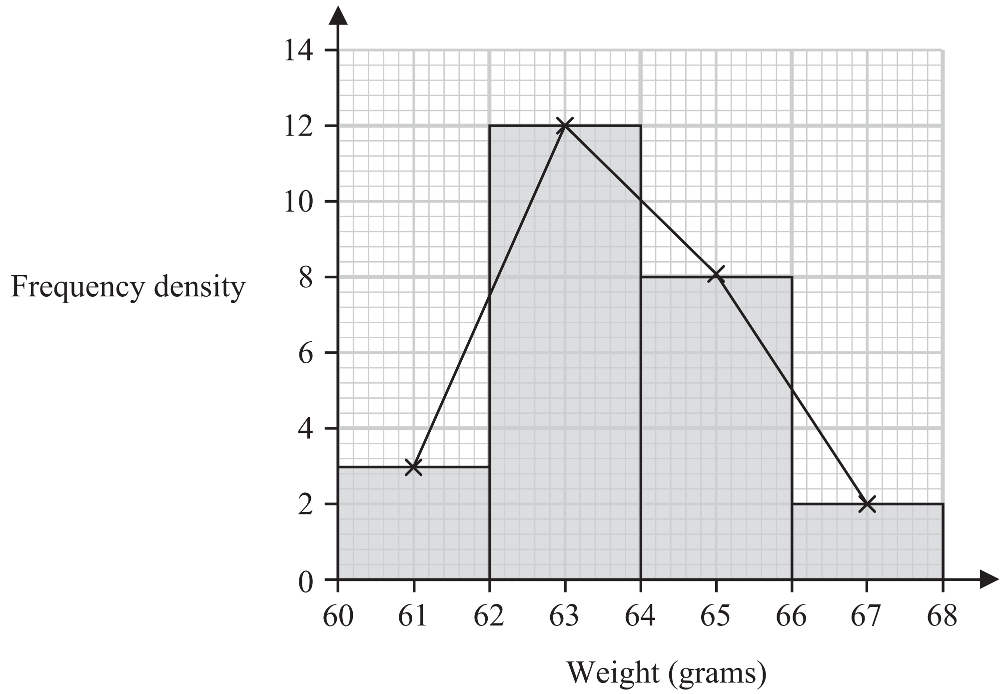
1. The histogram and its frequency polygon below give information about the weights, in grams, of 50 plums.



(a) Show that an estimate for the mean weight of the 50 plums is 63.72 grams.

(2)

(b) Calculate an estimate for the standard deviation of the 50 plums.

(2)

Later it was discovered that the scales used to weigh the plums were broken.

Each plum actually weighs 5 grams less than originally thought.

(c) State the effect this will have on the estimate of the standard deviation in part (b). Give a reason for your answer.

(1)

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Question 1 continued

(Total for Question 1 is 5 marks)

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2. Fred and Nadine are investigating whether there is a linear relationship between Daily Mean Pressure, p hPa, and Daily Mean Air Temperature, t °C, in Beijing using the 2015 data from the large data set.

Fred randomly selects one month from the data set and draws the scatter diagram in Figure 1 using the data from that month.

The scale has been left off the horizontal axis.

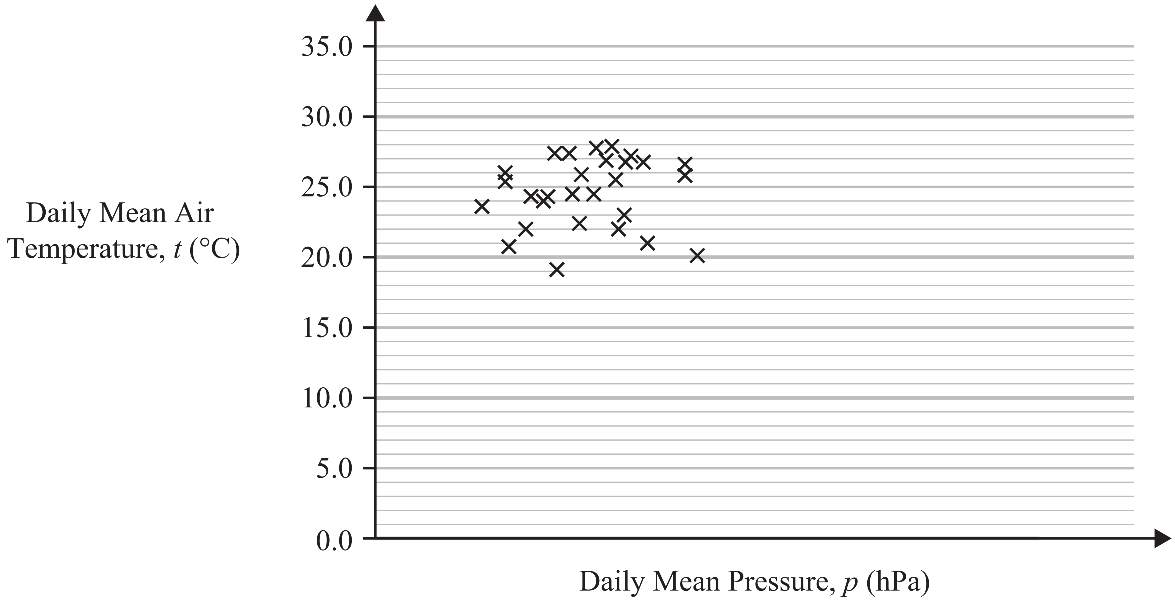

Figure 1

(a) Describe the correlation shown in Figure 1.

(1)

Nadine chooses to use all of the data for Beijing from 2015 and draws the scatter diagram in Figure 2.

She uses the same scales as Fred.

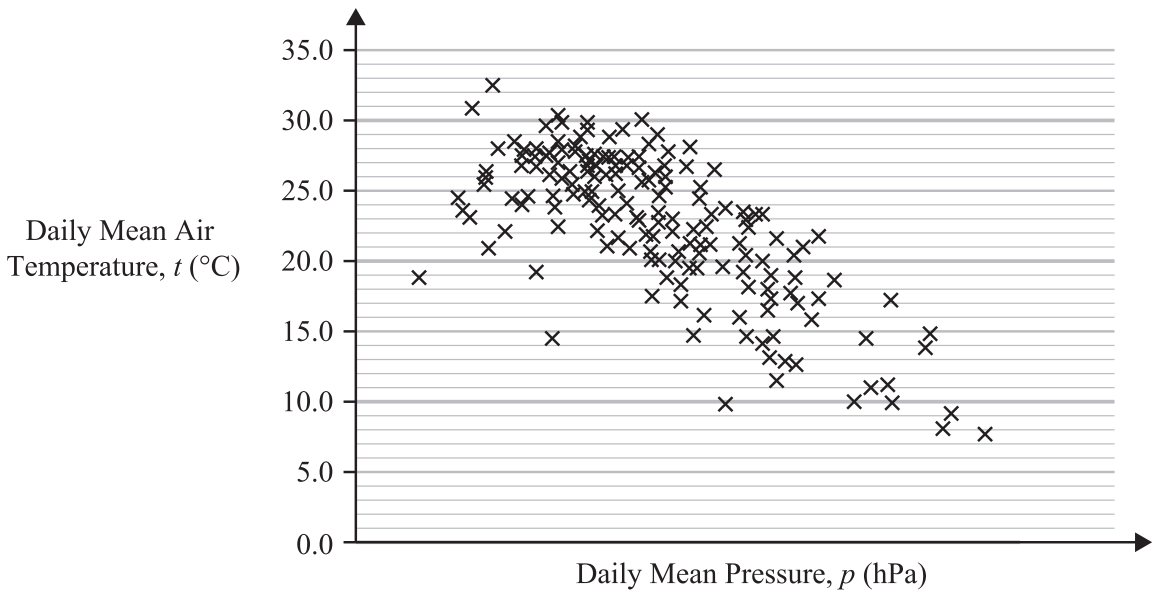

Figure 2

P 7 2 8 1 8 A 0 4 1 2

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Question 2 continued

(b) Explain, in context, what Nadine can infer about the relationship between p and t using the information shown in Figure 2.

(1)

(c) Using your knowledge of the large data set, state a value of p for which interpolation can be used with Figure 2 to predict a value of t.

(1)

(d) Using your knowledge of the large data set, explain why it is not meaningful to look for a linear relationship between Daily Mean Wind Speed (Beaufort Conversion) and Daily Mean Air Temperature in Beijing in 2015.

(1)

(Total for Question 2 is 4 marks)

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3. In an after-school club, students can choose to take part in Art, Music, both or neither.

There are 45 students that attend the after-school club. Of these

- 25 students take part in Art
- 12 students take part in both Art and Music
- the number of students that take part in Music is x

(a) Find the range of possible values of x

(2)

One of the 45 students is selected at random.

Event A is the event that the student selected takes part in Art.

Event M is the event that the student selected takes part in Music.

(b) Determine whether or not it is possible for the events A and M to be independent.

(4)

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Question 3 continued

(Total for Question 3 is 6 marks)

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4. Past information shows that 25% of adults in a large population have a particular allergy.

Rylan believes that the proportion that has the allergy differs from 25%

He takes a random sample of 50 adults from the population.

Rylan carries out a test of the null hypothesis H_0: $p = 0.25$ using a 5% level of significance.

(a) Write down the alternative hypothesis for Rylan's test.

(1)

(b) Find the critical region for this test.
You should state the probability associated with each tail, which should be as close to 2.5% as possible.

(4)

(c) State the actual probability of incorrectly rejecting H_0 for this test.

(1)

Rylan finds that 10 of the adults in his sample have the allergy.

(d) State the conclusion of Rylan's hypothesis test.

(1)

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Question 4 continued

(Total for Question 4 is 7 marks)

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P 7 2 8 1 8 A 0 9 1 2

Turn over

5. Julia selects 3 letters at random, one at a time without replacement, from the word

VARIANCE

The discrete random variable X represents the number of times she selects a letter A.

(a) Find the complete probability distribution of X.

(5)

Yuki selects 10 letters at random, one at a time **with** replacement, from the word

DEVIATION

(b) Find the probability that he selects the letter E at least 4 times.

(3)

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 8 marks)

TOTAL FOR STATISTICS IS 30 MARKS

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Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Thursday 23 May 2024

Afternoon

Paper reference **8MA0/21**

Mathematics

Advanced Subsidiary
PAPER 21: Statistics

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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Instructions

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Information

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- The total mark for this part of the examination is 30. There are 5 questions.
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Advice

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- Try to answer every question.
- If you change your mind about an answer, cross it out and put your new answer and any working underneath.
- Check your answers if you have time at the end.

Turn over

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Pearson

1. A coach recorded the heights of some adult rugby players and found the following summary statistics.

Median = 1.85 m

Range = 0.28 m

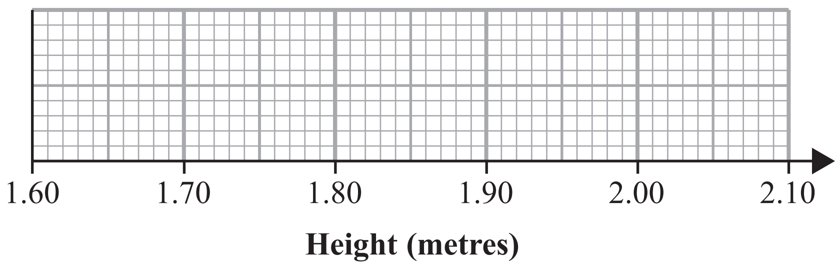
Interquartile range = 0.11 m

The coach also noticed that

- the height of the shortest player is 1.72 m
- 25% of the players' heights are below the height of a player whose height is 1.81 m

Draw a box and whisker plot to represent this information on the grid below.

(4)



Use the spare grid on page 3 if you need to redraw your box and whisker plot.

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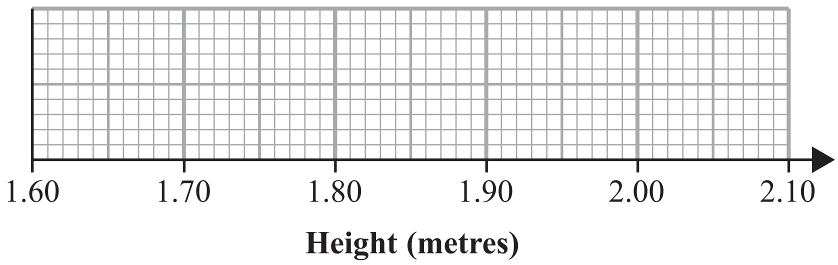
Question 1 continued

Only use this grid if you need to redraw your box and whisker plot.



(Total for Question 1 is 4 marks)

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2. Keith is studying the variable Daily Mean Wind Direction, in degrees, from the large data set.

Keith summarised the data for Camborne from 1987 into 4 directions A, B, C and D representing North, South, East and West in some order.

Direction	A	B	C	D
Frequency	22	48	56	58

(a) Using your knowledge of the large data set state, giving a reason, which direction A represents.

(1)

The entry for Hurn on 27th September 1987 was 999

(b) State, giving a reason, what Keith should do with this value.

(2)

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Question 2 continued

(Total for Question 2 is 3 marks)

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3. Customers in a shop have to queue to pay.

The partially completed table below and partially completed histogram opposite, give information about the time, x minutes, spent in the queue by each of 112 customers one day.

Time in queue (x minutes)	**Frequency**
1–2	64
2–3	
3–4	13
4–6	
6–8	3

No customer spent less than 1 minute or longer than 8 minutes in the queue.

(a) Complete the table.
(2)

(b) Complete the histogram.
(2)

Ting decides to model the **frequency density** for these 112 customers by a curve with equation

$$y = \frac{k}{x^2} \qquad 1 \leqslant x \leqslant 8$$

where k is a constant.

(c) Find the value of k
(3)

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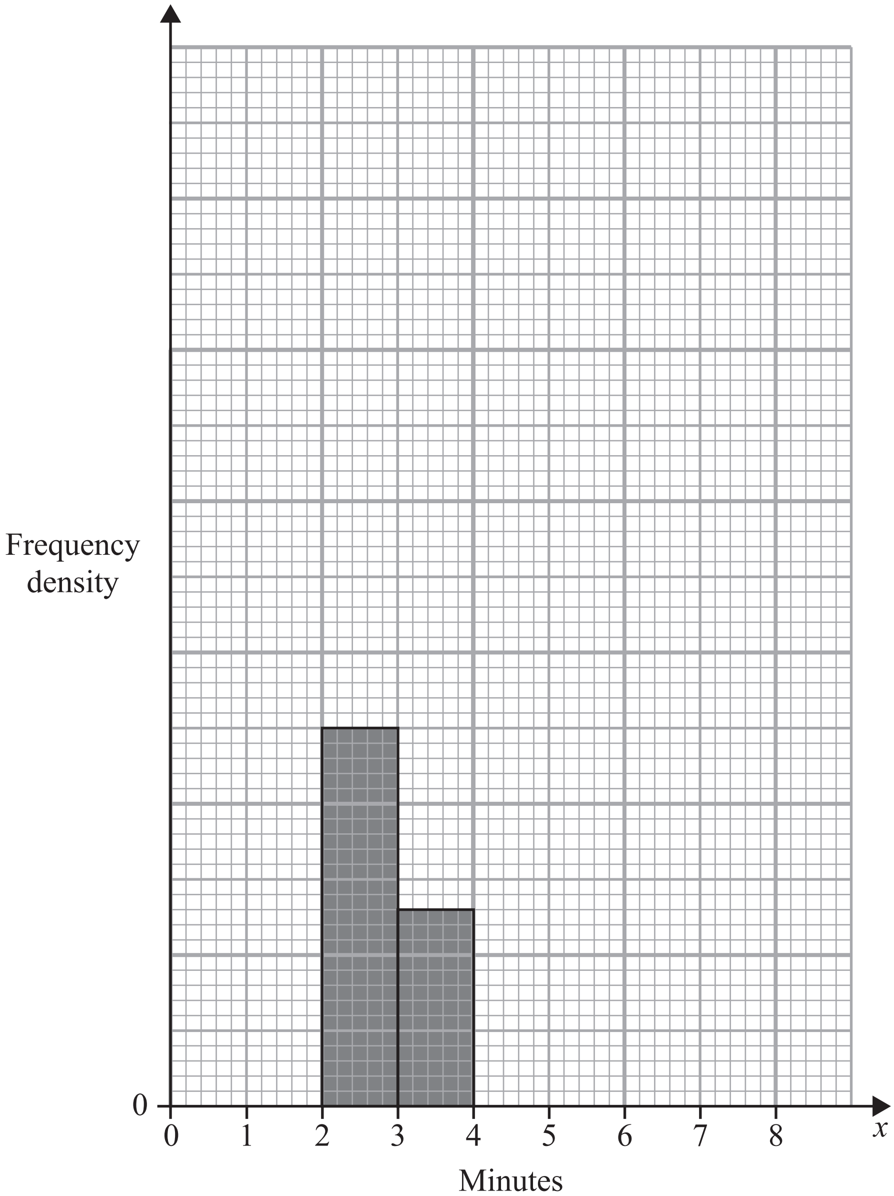
Question 3 continued



Turn over for a spare grid if you need to redraw your histogram.

Question 3 continued

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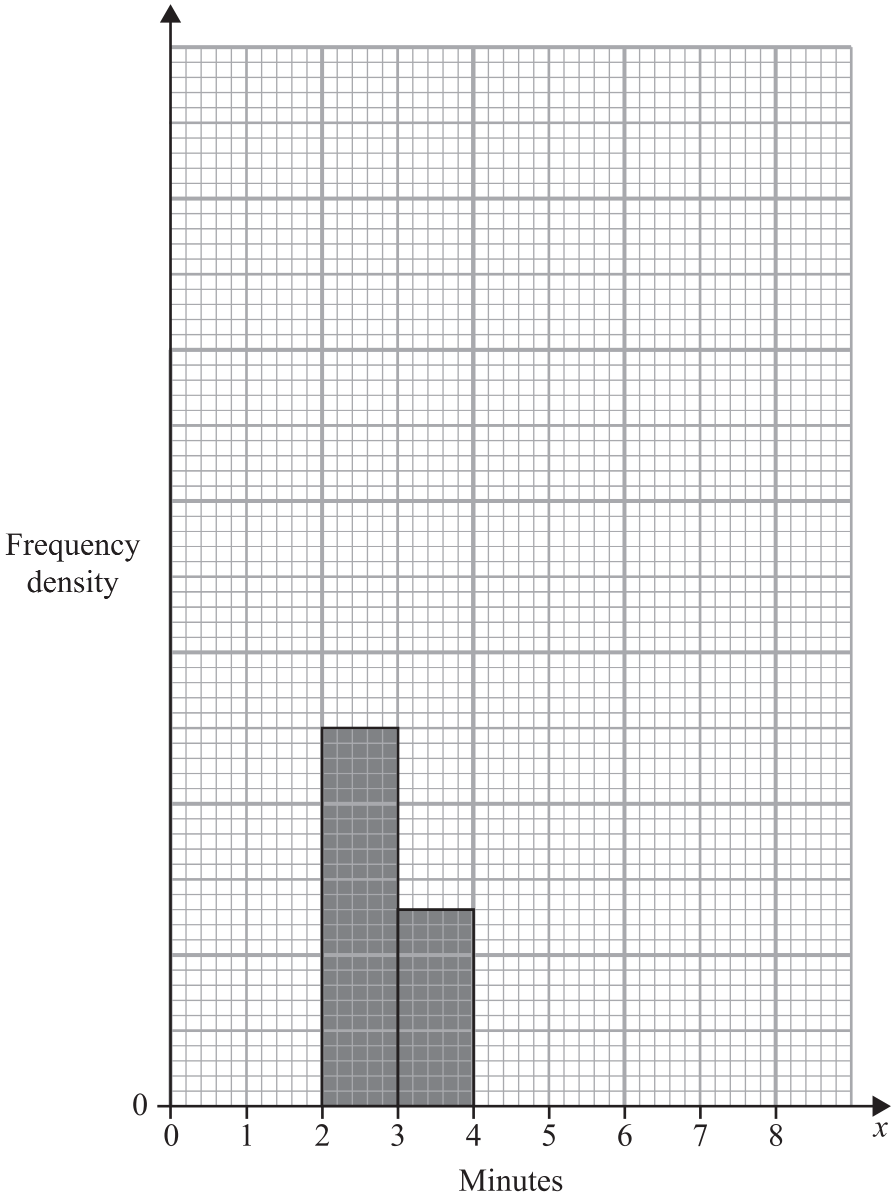
Question 3 continued

Only use this grid if you need to redraw your histogram.



(Total for Question 3 is 7 marks)

4. The random variable $X \sim B(27, 0.35)$

(a) Find (i) $P(X = 10)$

(ii) $P(12 \leqslant X < 15)$

(3)

Historical records show that the proportion of defective items produced by a machine is 0.12

Following a maintenance service of the machine, a random sample of 60 items is taken and 3 defective items are found.

(b) Carry out a suitable test to determine whether the proportion of defective items produced by the machine has decreased following the maintenance service.
You should state your hypotheses clearly and use a 5% level of significance.

(4)

(c) Write down the p-value for your test in part (b)

(1)

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Question 4 continued

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Question 4 continued

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Question 4 continued

(Total for Question 4 is 8 marks)

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5. A biased 4-sided spinner has the numbers 6, 7, 8 and 10 on it.

The discrete random variable X represents the score when the spinner is spun once and has the following probability distribution,

x	6	7	8	10
$P(X = x)$	0.5	0.2	q	q

where q is a probability.

(a) Find the value of q

(1)

Karen spins the spinner repeatedly until she **either** gets a 7 **or** she has taken 4 spins.

(b) Show that the probability that Karen stops after taking her 3rd spin is 0.128

(2)

The random variable S represents the number of spins Karen takes.

(c) Find the probability distribution for S

(4)

The random variable N represents the number of times Karen gets a 7

(d) Find $P(S > N)$

(1)

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 8 marks)

TOTAL FOR STATISTICS IS 30 MARKS

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Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Friday 23 May 2025

Afternoon

Paper reference **8MA0/21**

Mathematics

Advanced Subsidiary

PAPER 21: Statistics

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- Answer the questions in the spaces provided
  – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Values from statistical tables should be quoted in full. If a calculator is used instead of tables the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

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- The total mark for this part of the examination is 30. There are 5 questions.
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  – *use this as a guide as to how much time to spend on each question.*

Advice

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- Try to answer every question.
- If you change your mind about an answer, cross it out and put your new answer and any working underneath.
- Check your answers if you have time at the end.

Turn over ▶

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P 7 6 3 7 7 A 0 1 1 2

Pearson

1. The random variable $X \sim B(20, 0.37)$

(a) Find $P(X = 8)$

(1)

(b) Find $P(X \leqslant 5)$

(1)

(c) Find $P(X \leqslant 5.7)$

(1)

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Question 1 continued

(Total for Question 1 is 3 marks)

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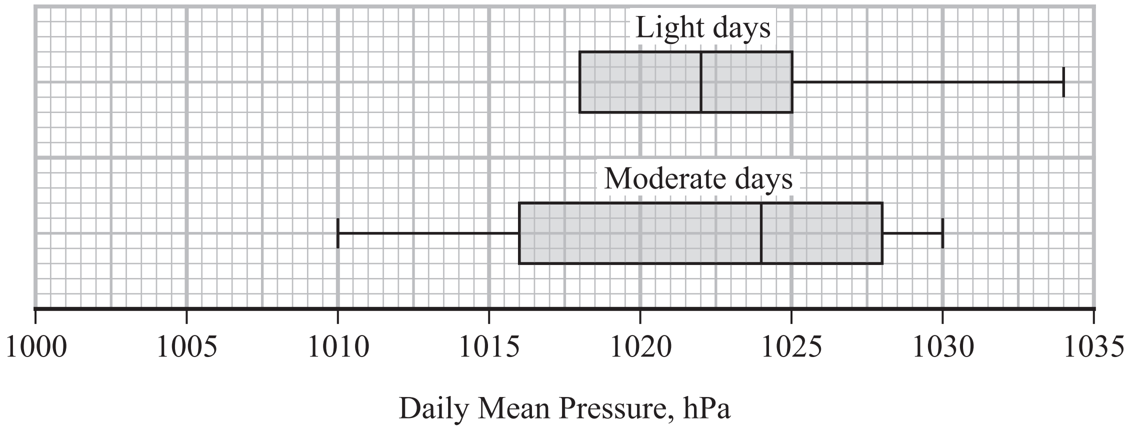
2. Jasper is investigating the relationship between Daily Mean Pressure and Daily Mean Windspeed (Beaufort conversion) for Perth in 2015 using the data from the large data set.
Treating the large data set for Perth in 2015 as the population, Jasper decides to use all the data available.

(a) Write down the name given to this method of data collection.

(1)

All of the Daily Mean Windspeed data for Perth in 2015 are classed as either "Light" or "Moderate".

Jasper splits the data for Perth in 2015 into Light days and Moderate days and draws a box plot for each set of data, but omits the lower tail for Light days.



The smallest three values of Daily Mean Pressure for Light days for Perth in 2015 are 1007, 1009 and 1010 hPa.

An outlier in the first quartile is defined as any value more than $1.5 \times \text{IQR}$ below Q_1

(b) (i) Determine if there are any outliers in the first quartile of Daily Mean Pressure for Light days.

(2)

(ii) Hence, complete the box plot of Daily Mean Pressure for Light days.

(1)

The box plot for Light days is based on data for 161 days.

(c) Using your knowledge of the large data set, estimate the number of Moderate days with a Daily Mean Pressure of 1028 hPa and higher in the large data set for Perth in 2015.

(2)

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Question 2 continued

(Total for Question 2 is 6 marks)

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3. A bag contains red counters, green counters and white counters only.
The table shows the proportion of each colour of counter that Elsa believes to be in the bag.

Colour	red	green	white
Proportion	p	0.2	$4p$

Elsa selects at random 40 counters from the bag, one at a time, with replacement.

Assuming Elsa's belief is true,

(a) find the distribution of the number of red counters Elsa selects.

(3)

Jayda believes that the true proportion of green counters in the bag is greater than 0.2

She takes a random sample of 40 counters from the bag, one at a time, with replacement.

There are 11 green counters in her sample.

(b) (i) Use a suitable test to assess Jayda's belief.

You should

- state your hypotheses clearly
- use a 5% level of significance
- state the p-value for the test

(4)

(ii) Find the acceptance region for the test in part (i).

(2)

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Question 3 continued

(Total for Question 3 is 9 marks)

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4. Giovanni believes there is a relationship between the mass of a car, m kg, and its fuel consumption, c miles per gallon (mpg).
For a sample of 25 cars, he obtains the following summary statistic.

$$\sum (c-\bar{c})^2 = 394$$

(a) Find the standard deviation of the fuel consumption of the 25 cars.

(1)

Using m as the explanatory variable, Giovanni creates the linear regression model

$$c = a + bm$$

where a and b are constants.

Using his model, he concludes that, on average

- the fuel consumption is 3.5 mpg lower for each additional 500 kg of mass
- the fuel consumption of a car with a mass of 1700 kg is 20 mpg

(b) (i) Find the value of b

(2)

(ii) Find the value of a

(2)

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Question 4 continued

(Total for Question 4 is 5 marks)

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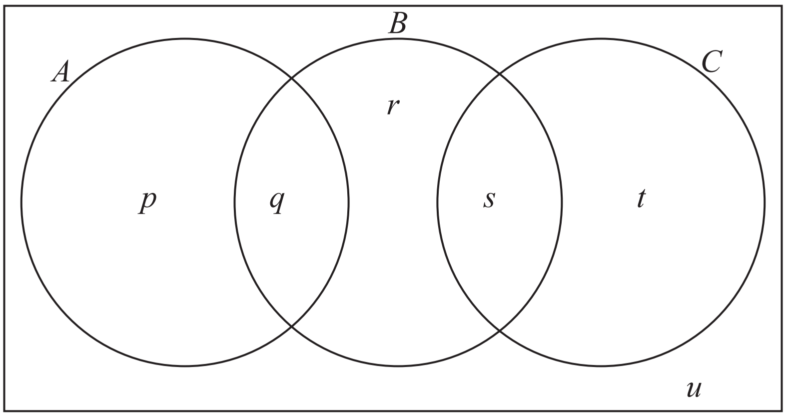
5. Events A, B and C are such that

- P(A or C) = 0.55
- the probability that none of the events occur is 0.23
- the probability that exactly two of the events occur is 0.18
- A and B are independent
- P(B) = P(A) + 0.1

The Venn diagram represents the events A, B and C and their associated probabilities p, q, r, s, t and u



(a) Write down the value of u

(1)

(b) Find the value of r

(2)

(c) Find

(i) the value of p

(ii) the value of q

(iii) the value of s

(iv) the value of t

(4)

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 7 marks)

TOTAL FOR STATISTICS IS 30 MARKS

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Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number | Candidate Number

Mock Paper Set 1

Paper Reference **8MA0/21**

Mathematics
Advanced Subsidiary
Paper 21: Statistics

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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Turn over ▶

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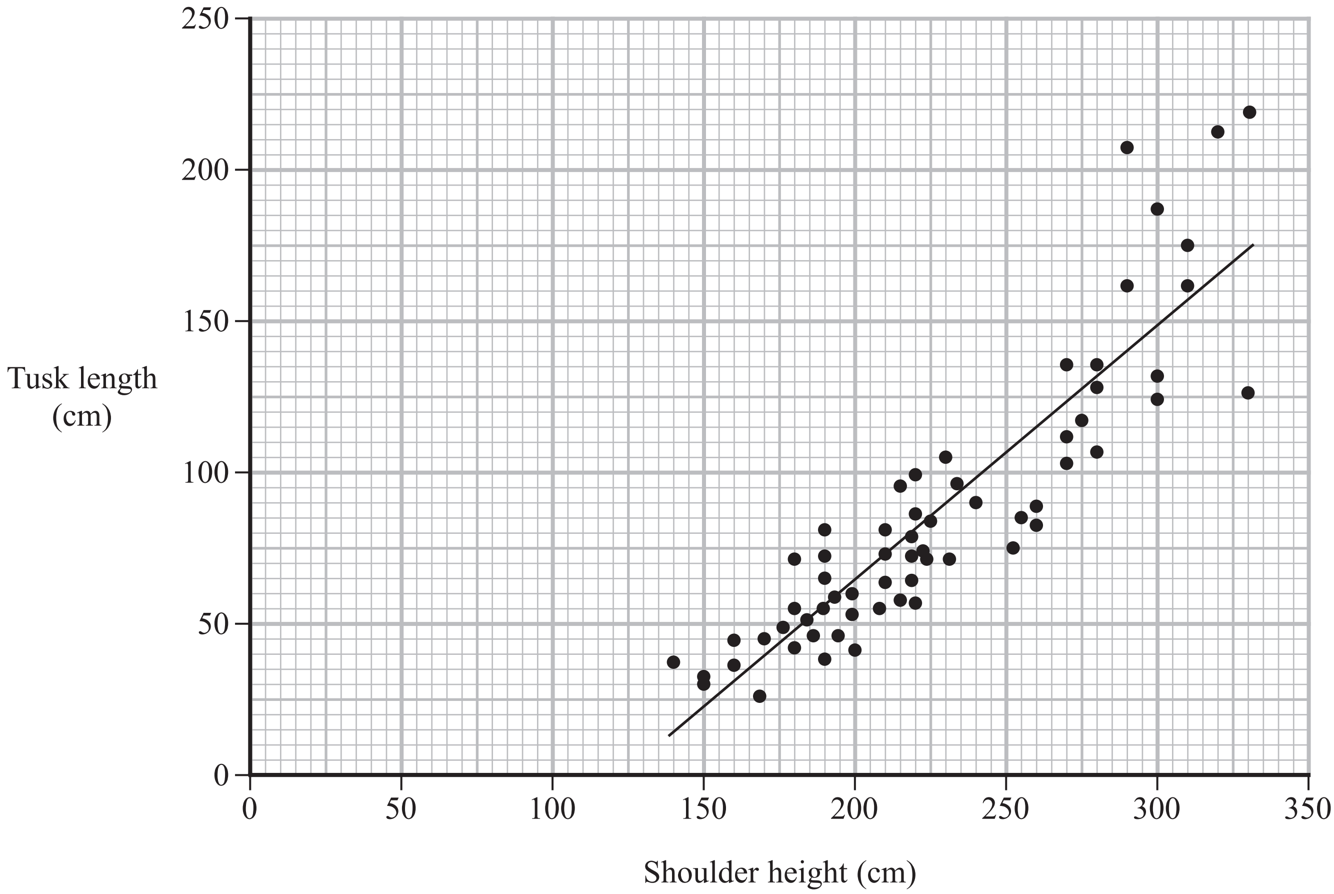
Pearson

1. Scientists collected data for some African elephants.

Matthew is studying the relationship between elephant shoulder height, s cm, and tusk length, t cm.

He uses only the data for male elephants and draws the following scatter diagram using statistical software.



(a) Give an interpretation of the correlation between tusk length and shoulder height.

(1)

The equation of the regression line of t on s for these data is $t = 0.843s - 104$

(b) Give an interpretation of the gradient of this regression line.

(1)

Matthew suggests using this regression line to draw conclusions about the relationship between elephant shoulder height and tusk length for **all** elephants.

(c) Explain why his suggestion may not be appropriate.

(1)

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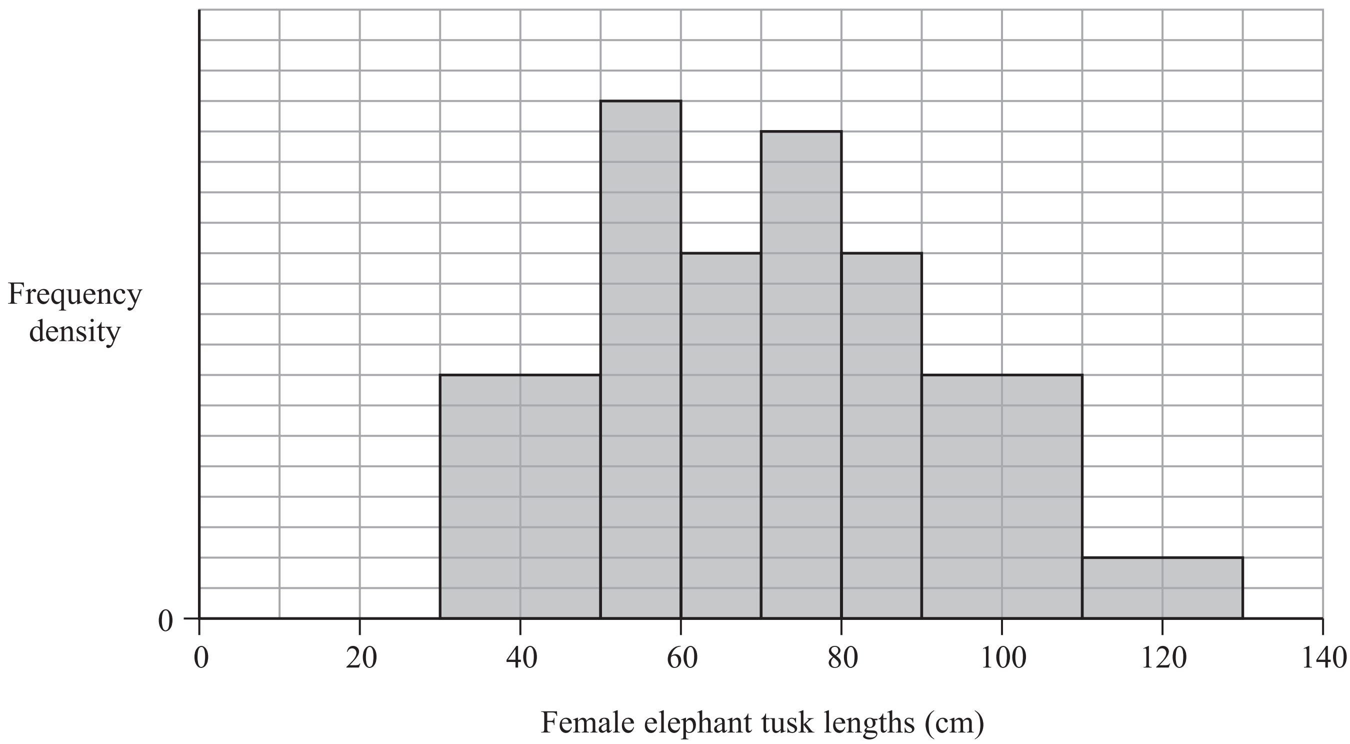
Question 1 continued

Matthew also draws the following histogram for the tusk lengths of female elephants.



(d) Estimate the percentage of female elephants with a tusk length of 75 cm to 100 cm.

(3)

Question 1 continued

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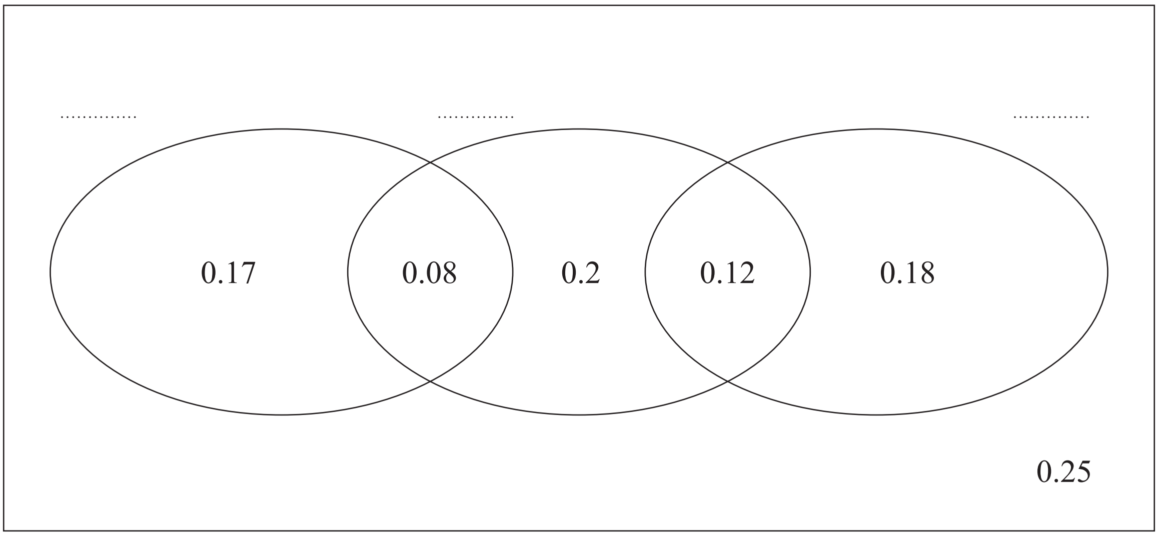
Question 1 continued

(Total for Question 1 is 6 marks)

2. The Venn diagram shows three events and the probabilities associated with each region.

The labels for the three events are missing.



The three events shown by the Venn diagram are A, B and C.

- Event A and event B are mutually exclusive
- Event A and event C are independent

(a) Giving reasons and showing your working, complete the Venn diagram with the correct labels for A, B and C.

(4)

(b) Write down the probability of event B not occurring.

(1)

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Question 2 continued

(Total for Question 2 is 5 marks)

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3. Kathleen is investigating the weather for Heathrow from May to October 2015.

She uses the large data set and produces the summary statistics below for the daily mean temperature (t°C).

$$n = 184 \qquad \sum t = 2876.7 \qquad \sum t^2 = 46\,797.3 \qquad \text{Maximum} = 28.7$$

Kathleen defines limits for outliers for each variable using the mean ± 3 times standard deviation rule.

(a) Show that $t = 28.7$ is an outlier.

(3)

Kathleen plans to produce summary statistics for the other weather variables recorded for Heathrow.

(b) Giving a reason for your answer, state a numerical weather variable from the large data set for which her outlier rule will not give practically useful results.

(2)

Kathleen summarised the daily mean visibility for Heathrow from May to October 2015 in the table below.

Daily mean visibility (x decametres)	**Number of days**
$400 \leqslant x < 1200$	7
$1200 \leqslant x < 2000$	24
$2000 \leqslant x < 2400$	32
$2400 \leqslant x < 2800$	40
$2800 \leqslant x < 3200$	30
$3200 \leqslant x < 3600$	26
$3600 \leqslant x < 4400$	25

(c) Use linear interpolation to find an estimate of the lower quartile.

(2)

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Question 3 continued

Another student, Mason, calculated the mean and standard deviation of the daily mean temperature (°F) in Jacksonville from May to October 2015.

Mean = 76.7 °F   Standard deviation = 4.86 °F

The formula to convert C degrees Celsius to F degrees Fahrenheit is $F = 1.8C + 32$

(d) Find, in degrees Celsius,

(i) the mean

(ii) the standard deviation

of the daily mean temperature in Jacksonville from May to October 2015.

(2)

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Question 3 continued

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Question 3 continued

(Total for Question 3 is 9 marks)

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4. A company produces blue beads and green beads.

The proportion of green beads produced is 0.38

Beads are selected at random and put into mixed packs of 10

(a) Find the probability of a pack containing more than 6 green beads.

(2)

(b) Find the most likely number of green beads in a pack.
Show your working clearly.

(2)

The company also produces bowls.

Initial tests suggest that 20% of these bowls are faulty.

The manager of the company believes that the proportion of faulty bowls is greater than 20%

He takes a random sample of 40 bowls.

(c) Using a 5% level of significance, find the critical region for a one-tailed test to check the manager's belief.
State your hypotheses clearly.

(3)

(d) Find the actual level of significance for this test.

(1)

The manager finds that 17 of the 40 bowls sampled are faulty.

(e) With reference to this information and part (c), comment on the manager's belief.

(1)

(f) Comment on the validity of the statistical model used in part (c), giving a reason for your answer.

(1)

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Question 4 continued

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Question 4 continued

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Question 4 continued

(Total for Question 4 is 10 marks)

TOTAL FOR STATISTICS IS 30 MARKS

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Pearson Edexcel Level 3 GCE

Paper reference **8MA0/21**

Mathematics

Advanced Subsidiary
PAPER 21: Statistics
Mock Set 2

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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Turn over ▶

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Pearson

1. A sample of independent values is taken from a discrete random variable X

The sample data are summarised in the grouped frequency table below

x	2 or 3	4	5	6	7	8, 9 or 10
Frequency	20	40	$95 - n$	n	25	15

Outliers for x are defined as any values

$$\text{below } Q_1 - 1.5 \times \text{IQR} \qquad \text{or} \qquad \text{above } Q_3 + 1.5 \times \text{IQR}$$

Given that in these data, $Q_1 = 4$ and $Q_3 = 6$

(a) determine whether there may be outliers in these data.
Give a reason for your answer.

(2)

Given that the median of these data is 5

(b) find the maximum value of n

(3)

A value of x is chosen at random from the sample.

(c) Find the probability that this value is greater than 7

(1)

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Question 1 continued

(Total for Question 1 is 6 marks)

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2. Bea believes that 20% of letters are sent by 1st class post.
Marco claims that less than 20% of letters are sent by 1st class post.

Marco and Bea decide to carry out an appropriate hypothesis test using a 5% significance level.

(a) Write down appropriate null and alternative hypotheses for their test.
(1)

Marco and Bea take a random selection of 25 letters and find that 1 letter was sent by 1st class post.

(b) (i) Showing your working, find the critical region for their test.

(ii) State the conclusion of their test.
(4)

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Question 2 continued

(Total for Question 2 is 5 marks)

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3. Edward is practising statistical calculations using values from the large data set.

He selects a random day from June 1987 and calculates the following summary statistics for the Daily Mean Temperature, x °C, for that day from the 8 locations in the large data set.

$$S_{xx} = 133 \qquad \sum x^2 = 2311$$

(a) Using these statistics,

(i) show that the mean of x is 16.5

(3)

(ii) find the standard deviation of x

(2)

Edward now studies the Daily Mean Visibility, v, for the same day in June 1987

He codes the data for 5 locations in the large data set using

$$y = \frac{v}{100} - 20$$

and finds that the mean of y is 10.6 and the standard deviation of y is 12.6

(b) (i) Show that the mean of v is 3060

(2)

(ii) Find the standard deviation of v

(1)

(c) Using your knowledge of the large data set,

(i) state the units for v

(1)

(ii) state, giving a reason, a location in the large data set that Edward could **not** use for his Daily Mean Visibility calculations.

(1)

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Question 3 continued

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Question 3 continued

(Total for Question 3 is 10 marks)

4. A sports club offers its members squash, tennis and a multi-gym.

The club secretary carries out a census to discover how many members play squash, play tennis or use the multi-gym.

(a) Explain what you understand by a census in this context.

(1)

For a randomly selected member of the club, the events S, T and G are defined as

S, the club member plays squash

T, the club member plays tennis

G, the club member uses the multi-gym

The club secretary finds that S and G are independent events.

Given that

$$\mathrm{P}(S) = \frac{3}{4} \qquad \mathrm{P}(T) = \frac{3}{10} \qquad \mathrm{P}(G) = \frac{1}{5}$$

(b) show that for a randomly selected member of the club

$$\mathrm{P}(S \text{ and } G) = 0.15$$

(2)

The club secretary also finds that

- all club members take part in at least one of these three activities
- T and G are mutually exclusive events

(c) (i) Showing your working clearly, find the probability that a randomly selected member of the club plays squash but does **not** play tennis and does **not** use the multi-gym.

(2)

(ii) Draw a Venn diagram to display the three events S, T and G and all of their associated probabilities.

(4)

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Question 4 continued

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Question 4 continued

(Total for Question 4 is 9 marks)

TOTAL FOR PAPER IS 30 MARKS

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Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Mock Paper Set 3

Paper reference **8MA0/21**

Mathematics

Advanced Subsidiary

PAPER 21: Statistics

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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Turn over

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Pearson

1 A boat company organises a trip to see whales each day.
The company advertises that customers see whales on 80% of its trips.
Throughout the year, Ali goes on 10 of these trips on randomly chosen days.

(a) Suggest a suitable distribution to model the number of trips where Ali sees whales.
(1)

(b) Using this model, find the probability that Ali sees whales on

(i) exactly 8 of the trips,
(ii) at least 7 of the trips.
(3)

Sam really wants to see whales, so goes on a trip each day for 10 consecutive days.

(c) State a limitation of using the model in part (a) to represent the number of trips where Sam sees whales.
(1)

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Question 1 continued

(Total for Question 1 is 5 marks)

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2 Amelie is a basketball player.
In one season, she scores from 40% of her free throws.

At the end of the season her sports psychologist retires.

Before the start of the next season, Amelie trains with a new sports psychologist.
After training with the new sports psychologist, Amelie takes 60 free throws and scores from 31 of them.

Amelie believes that training with the new sports psychologist has changed the proportion of her free throws from which she scores.

(a) Using a suitable test, assess Amelie's belief.

You should

- state your hypotheses and conclusion clearly
- use a 5% significance level

(4)

(b) Find the p-value for the test in part (a).

(1)

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Question 2 continued

(Total for Question 2 is 5 marks)

3 A machine produces bolts for ceiling fans.
The machine is set to produce bolts of length 26 mm.
For a bolt to be usable, its length must be (26 ± 2) mm.

A sample of 100 bolts was taken.
The bolts were measured and the results summarised in the table below

Length of bolt (b mm)	**Midpoint (x mm)**	**Frequency**
$21 < b \leqslant 22$	21.5	1
$22 < b \leqslant 23$	22.5	5
$23 < b \leqslant 24$	23.5	8
$24 < b \leqslant 25$	24.5	15
$25 < b \leqslant 26$	25.5	31
$26 < b \leqslant 27$	26.5	24
$27 < b \leqslant 28$	27.5	12
$28 < b \leqslant 29$	28.5	4

(a) Estimate the proportion of bolts in this sample that are usable.
(1)

(b) Use linear interpolation to find an estimate for the median bolt length.
(2)

Given that $\sum x = 2560 \quad \sum x^2 = 65\ 751$

(c) find an estimate for

(i) the mean bolt length,
(ii) the standard deviation of the bolt lengths.
(3)

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Question 3 continued

(Total for Question 3 is 6 marks)

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4 Huan plays a game using a fair 8-sided spinner, numbered 1 to 8
Huan spins the spinner until the game stops.

The game stops according to the following rules

- on the first spin, if the spinner lands on a **square** number, the game stops
- on the second spin, if the spinner lands on an **even** number, the game stops
- on the third spin, if the spinner lands on a **prime** number, the game stops
- if the game is still going after 3 spins, one final spin is taken and the game stops

The random variable S represents the number of times the spinner is spun until the game stops.

(a) Show that $P(S = 2) = \frac{3}{8}$

(2)

(b) Complete the probability distribution table below for the random variable S

(2)

s	1	2	3	4
$P(S = s)$		$\frac{3}{8}$		

(c) Find $P(1.6 < S < 3.2)$

(1)

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Question 4 continued

(Total for Question 4 is 5 marks)

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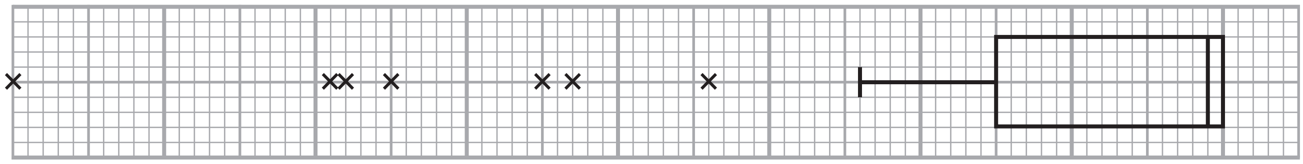

Luke finds the diagram above, showing a box plot with no axis.
The diagram lies next to a set of data labelled 'Large data set: Daily Total Rainfall – Hurn 1987'.

Luke thinks that the diagram should be rotated through 180° for it to represent these data.

(a) Using your knowledge of the large data set, explain why Luke is correct.

(1)

A sample of data for the Daily Mean Visibility in Leeming in 1987 was taken and is summarised in the table below

Lower Quartile (dam)	Upper Quartile (dam)
1000	x

An outlier is defined as any value
more than 1.5 × the interquartile range **above** the upper quartile
or more than 1.5 × the interquartile range **below** the lower quartile

The last 6 values of the ordered data are 3300, 3400, 3400, 3600, 3600, 3700

Luke believes that there are 5 outliers at the upper end of the data.

(b) Find the range of possible values for x if Luke is correct.

(4)

The first 6 values of the ordered data are 200, 200, 300, 400, 400, 600

Nish believes that there are 5 outliers at the lower end of the data, which gives the range of possible values for x as $1267 \leqslant x < 1400$

Given that there are only 5 outliers **in total** in the raw data

(c) explain why Nish's belief is **not** correct.

(2)

Luke wants to investigate the Daily Mean Windspeed (Beaufort conversion) from the large data set for Leeming in 2015.

He plans to select data until he has 25 pieces of data from **each** of the three categories of windspeed (light, moderate and fresh).

(d) State the sampling technique Luke plans to use.

(1)

(e) Using your knowledge of the large data set, explain clearly why Luke's sampling technique may **not** be successful in obtaining his sample.

(1)

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Question 5 continued

(Total for Question 5 is 9 marks)

TOTAL FOR STATISTICS IS 30 MARKS

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Pearson Edexcel Level 3 GCE	
Mathematics **Advanced Subsidiary** **Paper 2: Statistics and Mechanics**	
Sample assessment material for first teaching September 2017 **Time: 1 hour 15 minutes**	Paper Reference(s) **8MA0/02**
You must have: Mathematical Formulae and Statistical Tables Calculator	

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions
- Use **black** ink or ball-point pen
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- There are **two** sections in this question paper. Answer **all** the questions in Section A and **all** the questions in Section B.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information
- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 9 questions in this question paper. The total mark for this paper is 60.
- The marks for each question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice
- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end
- If you change your mind about an answer, cross it out and put your new answer and any working underneath.

SECTION A: STATISTICS

Answer ALL questions. Write your answers in the spaces provided.

1. Sara is investigating the variation in daily maximum gust, t kn, for Camborne in June and July 1987.

She used the large data set to select a sample of size 20 from the June and July data for 1987. Sara selected the first value using a random number from 1 to 4 and then selected every third value after that.

(a) State the sampling technique Sara used.

(1)

(b) From your knowledge of the large data set, explain why this process may not generate a sample of size 20.

(1)

The data Sara collected are summarised as follows

$$n = 20 \qquad \sum t = 374 \qquad \sum t^2 = 7600$$

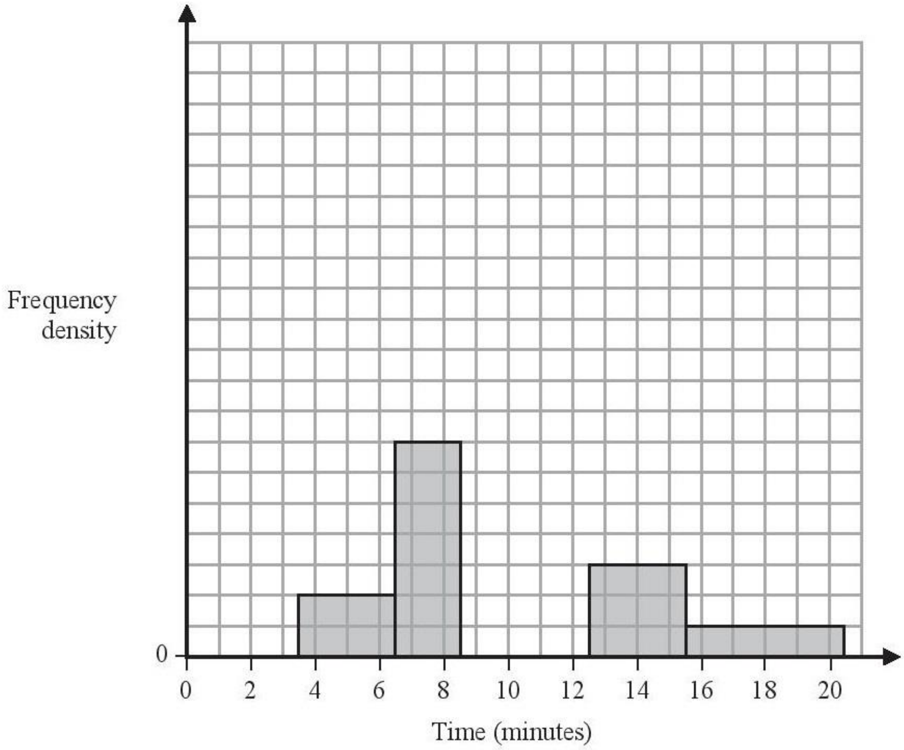
(c) Calculate the standard deviation.

(2)

Question 1 continued

(Total for Question 1 is 4 marks)

2. The partially completed histogram and the partially completed table show the time, to the nearest minute, that a random sample of motorists were delayed by roadworks on a stretch of motorway.



Delay (minutes)	Number of motorists
4 – 6	6
7 – 8	
9	17
10 – 12	45
13 – 15	9
16 – 20	

Estimate the percentage of these motorists who were delayed by the roadworks for between 8.5 and 13.5 minutes.

(5)

Question 2 continued

(Total for Question 2 is 5 marks)

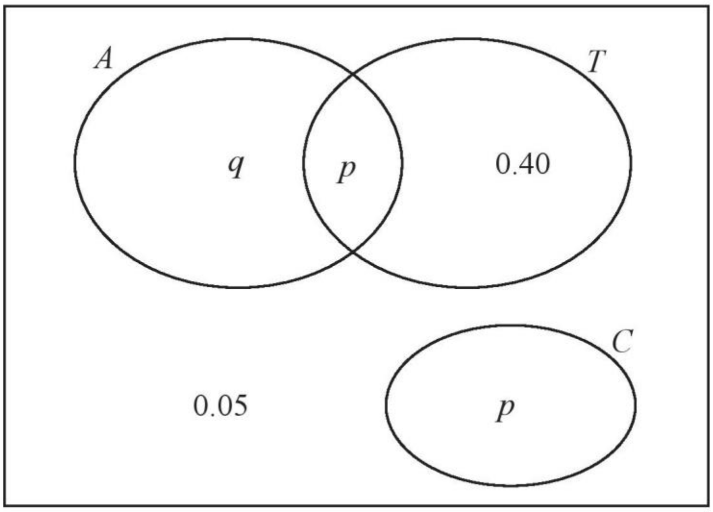
3. The Venn diagram shows the probabilities for students at a college taking part in various sports.

A represents the event that a student takes part in Athletics.

T represents the event that a student takes part in Tennis.

C represents the event that a student takes part in Cricket.

p and *q* are probabilities.



The probability that a student selected at random takes part in Athletics or Tennis is 0.75

(a) Find the value of *p*.

(1)

(b) State, giving a reason, whether or not the events *A* and *T* are statistically independent. Show your working clearly.

(3)

(c) Find the probability that a student selected at random does not take part in Athletics or Cricket.

(1)

Question 3 continued

(Total for Question 3 is 5 marks)

4. Sara was studying the relationship between rainfall, r mm, and humidity, h %, in the UK. She takes a random sample of 11 days from May 1987 for Leuchars from the large data set.

She obtained the following results.

h	93	86	95	97	86	94	97	97	87	97	86
r	1.1	0.3	3.7	20.6	0	0	2.4	1.1	0.1	0.9	0.1

Sara examined the rainfall figures and found

$$Q_1 = 0.1 \qquad Q_2 = 0.9 \qquad Q_3 = 2.4$$

A value that is more than 1.5 times the interquartile range (IQR) above Q_3 is called an outlier.

(a) Show that $r = 20.6$ is an outlier.

(1)

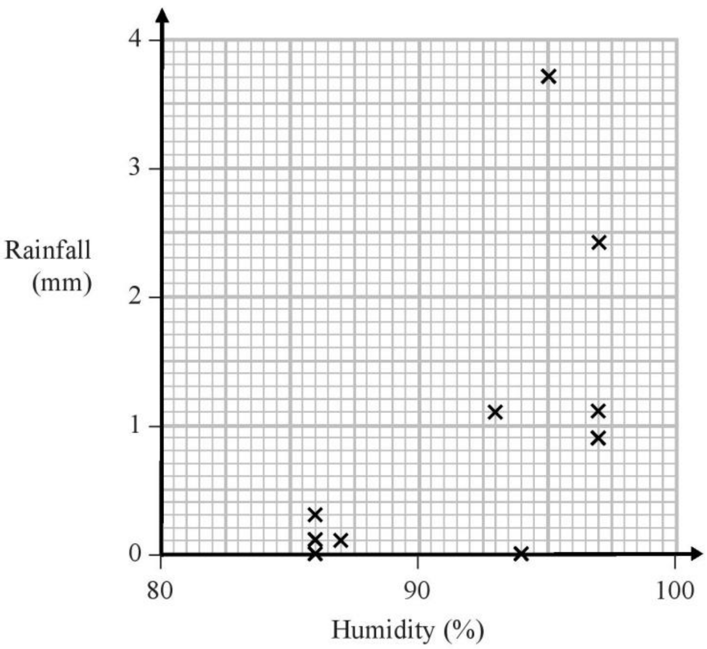
(b) Give a reason why Sara might (i) include

(ii) exclude

this day's reading.

(2)

Sara decided to exclude this day's reading and drew the following scatter diagram for the remaining 10 days' values of r and h.



(c) Give an interpretation of the correlation between rainfall and humidity.

(1)

Question 4 continued

The equation of the regression line of r on h for these 10 days is $r = -12.8 + 0.15h$

(d) Give an interpretation of the gradient of this regression line.

(1)

(e) (i) Comment on the suitability of Sara's sampling method for this study.

(ii) Suggest how Sara could make better use of the large data set for her study.

(2)

Total for Question 4 is 7 marks)

5. (a) The discrete random variable $X \sim B(40, 0.27)$

Find $P(X \geqslant 16)$

(2)

Past records suggest that 30% of customers who buy baked beans from a large supermarket buy them in single tins. A new manager suspects that there has been a change in the proportion of customers who buy baked beans in single tins. A random sample of 20 customers who had bought baked beans was taken.

(b) Write down the hypotheses that should be used to test the manager’s suspicion.

(1)

(c) Using a 10% level of significance, find the critical region for a two-tailed test to answer the manager's suspicion. You should state the probability of rejection in each tail, which should be less than 0.05

(3)

(d) Find the actual significance level of a test based on your critical region from part (c).

(1)

One afternoon the manager observes that 12 of the 20 customers who bought baked beans, bought their beans in single tins.

(e) Comment on the manager’s suspicion in the light of this observation.

(1)

Later it was discovered that the local scout group visited the supermarket that afternoon to buy food for their camping trip.

(f) Comment on the validity of the model used to obtain the answer to part (e), giving a reason for your answer.

(1)

Question 5 continued

(Total for Question 5 is 9 marks)

TOTAL FOR SECTION A IS 30 MARKS

Write your name here

Surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number	Candidate Number

Mathematics

Advanced Subsidiary
Paper 2: Statistics and Mechanics

Specimen Paper **Time: 1 hour 15 minutes**	Paper Reference **8MA0/02**

You must have: Mathematical Formulae and Statistical Tables, calculator	Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- There are **two** sections in this question paper. Answer **all** the questions in Section A and **all** the questions in Section B.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 60.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- If you change your mind about an answer, cross it out and put your new answer and any working underneath.

Turn over ▶

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SECTION A: STATISTICS

Answer ALL questions. Write your answers in the spaces provided.

1. A company manager is investigating the time taken, t minutes, to complete an aptitude test. The human resources manager produced the table below of coded times, x minutes, for a random sample of 30 applicants.

Coded time (x minutes)	**Frequency (f)**	**Coded time midpoint (y minutes)**
$0 \le x < 5$	3	2.5
$5 \le x < 10$	15	7.5
$10 \le x < 15$	2	12.5
$15 \le x < 25$	9	20
$25 \le x < 35$	1	30

(You may use $\sum fy = 355$ and $\sum fy^2 = 5675$)

(a) Use linear interpolation to estimate the median of the coded times.
(2)

(b) Estimate the standard deviation of the coded times.
(2)

The company manager is told by the human resources manager that he subtracted 15 from each of the times and then divided by 2, to calculate the coded times.

(c) Calculate an estimate for the median and the standard deviation of t.
(3)

The following year, the company has 25 positions available. The company manager decides not to offer a position to any applicant who takes 35 minutes or more to complete the aptitude test.

The company has 60 applicants.

(d) Comment on whether or not the company manager's decision will result in the company being able to fill the 25 positions available from these 60 applicants. Give a reason for your answer.
(2)

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Question 1 continued

(Total for Question 1 is 9 marks)

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2. The discrete random variable $X \sim B(30, 0.28)$

(a) Find $P(5 \leqslant X < 12)$

(2)

Past records from a large supermarket show that 25% of people who buy eggs, buy organic eggs. On one particular day, a random sample of 40 people is taken from those that had bought eggs and 16 people are found to have bought organic eggs.

(b) Test, at the 1% significance level, whether or not the proportion, p, of people who bought organic eggs that day had increased. State your hypotheses clearly.

(5)

(c) State the conclusion you would have reached if a 5% significance level had been used for this test.

(1)

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Question 2 continued

(Total for Question 2 is 8 marks)

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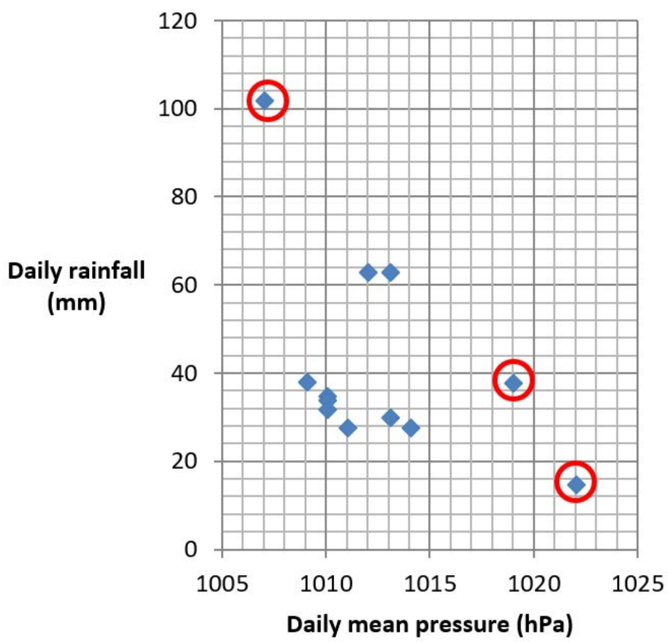
3. Pete is investigating the relationship between daily rainfall, w mm, and daily mean pressure, p hPa, in Perth during 2015. He used the large data set to take a sample of size 12. He obtained the following results.

p	1007	1012	1013	1009	1019	1010	1010	1010	1013	1011	1014	1022
w	102.0	63.0	63.0	38.4	38.0	35.0	34.2	32.0	30.4	28.0	28.0	15

Pete drew the following scatter diagram for the values of w and p and calculated the quartiles.

	Q_1	Q_2	Q_3
p	1010	1011.5	1013.5
w	29.2	34.6	50.7



An outlier is a value which is more than 1.5 times the interquartile range above Q3 or more than 1.5 times the interquartile range below Q1.

(a) Show that the 3 points circled on the scatter diagram above are outliers.

(2)

(b) Describe the effect of removing the 3 outliers on the correlation between daily rainfall and daily mean pressure in this sample.

(1)

John has also been studying the large data set and believes that the sample Pete has taken is not random.

(c) From your knowledge of the large data set, explain why Pete's sample is unlikely to be a random sample.

(1)

Question 3 continued

John finds that the equation of the regression line of w on p, using all the data in the large data set, is

$$w = 1023 - 0.223p$$

(d) Give an interpretation of the figure –0.223 in this regression line.

(1)

John decided to use the regression line to estimate the daily rainfall for a day in December when the daily mean pressure is 1011 hPa.

(e) Using your knowledge of the large data set, comment on the reliability of John's estimate.

(1)

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Question 3 continued

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Question 3 continued

(Total for Question 3 is 6 marks)

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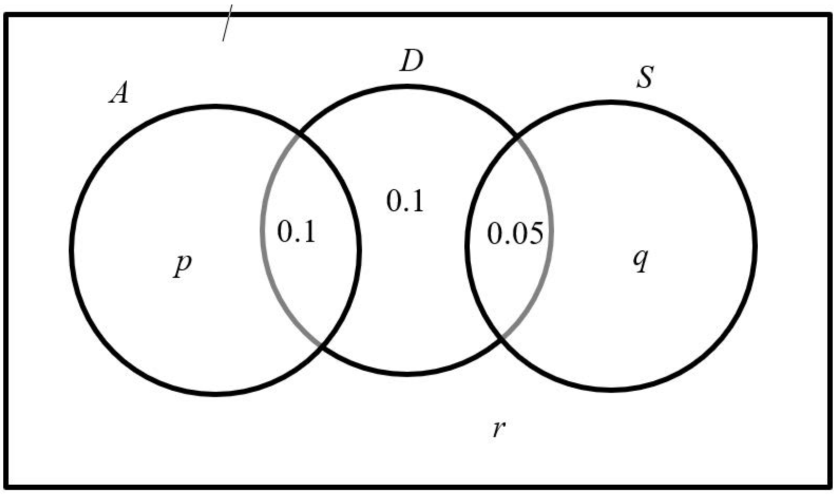
4. Alyona, Dawn and Sergei are sometimes late for school.
The events A, D and S are as follows

A Alyona is late for school

D Dawn is late for school

S Sergei is late for school

The Venn diagram below shows the three events *A*, *D* and *S* and the probabilities associated with each region of *D*. The constants *p*, *q* and *r* each represent probabilities associated with the three separate regions outside D.



(a) Write down 2 of the events *A*, *D* and *S* that are mutually exclusive. Give a reason for your answer.

(1)

The probability that Sergei is late for school is 0.2
The events *A* and *D* are independent.

(b) Find the value of *r*

(4)

Dawn and Sergei's teacher believes that when Sergei is late for school, Dawn tends to be late for school.

(c) State whether or not *D* and *S* are independent, giving a reason for your answer.

(1)

(d) Comment on the teacher's belief in the light of your answer to part (c).

(1)

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Question 4 continued

(Total for Question 4 is 7 marks)

TOTAL FOR SECTION A IS 30 MARKS